

Distribution of Work

Khush Sushil Poddar (24200202)

Khush led the integration of the large language model (LLM) and prompt engineering for the project. He implemented and iterated the `llm_service`, designed the prompt templates to include site metadata, time windows, and precomputed statistics, and introduced guardrails to ensure responses stayed grounded in actual measurements, including explicit numeric ranges and uncertainty statements. He stabilized the `/api/ask-llm` route, validated outputs against expert-written exemplars, and improved clarity by refining prompt scaffolds. To support performance and reliability, he monitored response timings and worked with the backend team to cap payload sizes for faster calls.

Beyond the LLM work, Khush also played a central role in building the project's data backbone. He migrated CSVs to PostgreSQL for deployment, extended the dataset with new sites and records, and implemented fuzzy site-name matching to align HAEDAT events with measurement data. He authored the Copernicus chlorophyll-a acquisition script, integrated HAEDAT, and enforced chunked loading in `data_service.py` to prevent timeouts. Additionally, he containerized the backend, conducted code cleanups, and wrote early project-plan documents to align roles and milestones across the team. He co-authored the written report (LLM integration, methodology, and evaluation sections) and co-led the final presentation, including slide development, demo narrative, and live walkthrough.

Shikhar Singh Negi (24203095)

Shikhar took ownership of backend orchestration and deployment. He standardized API paths, environment variable naming conventions, and pagination/query limits across `/api/measurements` and `/api/summary`, ensuring consistency and stability. He also managed backend pull requests, oversaw merges, and maintained interface reliability throughout development. Shikhar introduced deployment artefacts for Render, fixed route and index wiring issues after the database migration and authored the README and deployment notes so the repository could be cloned and run with minimal effort.

In addition to deployment, Shikhar worked extensively on performance and reliability. He optimized query limits while maintaining indexed access patterns, introduced error handling mechanisms, and implemented health-check logic for continuous service monitoring. He coordinated merges involving API patches, database migration, and dockerisation, and carried out load testing after each release to safeguard stability. His efforts ensured that the Flask layer acted as a predictable, low-latency contract between the frontend, the database, and the LLM service. He co-authored the report sections on system architecture, backend APIs, deployment, and performance, and presented the architecture portions during the final presentation.

Harshal Chandrakant More (24200354)

Harshal led the project's data sourcing and preprocessing efforts. He iterated the data service to support chunked ingestion and enforced schema consistency across chlorophyll-a, SST, and auxiliary variables, while tuning joins and filters for site/date windows. He designed and helped define materialized views to power efficient summary queries and contributed the architectural diagram for the system. Early on, he refined aspects of the LLM service to improve answer structure and stability, and he later worked closely with the frontend team to resolve naming and compatibility issues when enhanced datasets introduced new site records.

From an operational perspective, Harshal authored Dockerfiles to streamline local and cloud builds and documented data-loading steps for reproducibility. He carried out data quality passes to normalize timestamps and coordinates, handled missing values, and validated record counts after imports to prevent drift in analysis. He also captured meeting minutes and implementation decisions to maintain alignment across the team, and he provided fixes whenever ingestion or query assumptions shifted during development. He co-authored the report (data pipeline, sources, processing, and architecture figures) and presented the data ingestion/materialized-views workflow and system diagram in the final presentation.

Anuj Lalitkumar Lamba (24202506)

Anuj was responsible for developing the core frontend components using React 18 and integrating them with the project's APIs. He implemented the overview panel with search, filtering, date windows, record limits, and dataset summaries, alongside the site analytics view displaying key indicators and secondary variables. He developed interactive time-series charts for chlorophyll-a, SST, and bloom probability using Recharts, integrated the React Leaflet map, and built the measurement table with sorting and filtering functions. Anuj also enabled one-click CSV/JSON export and ensured each view respected user-selected date ranges.

His focus was on building a responsive, stable, and user-friendly dashboard. He worked on state management and error boundaries, so the system remained functional under large queries or transient network issues. Anuj coordinated closely with backend changes, updated components as APIs evolved, and ran UI regression testing after each merge. His contributions ensured that complex environmental analytics were presented in an intuitive, efficient, and accessible manner for non-expert users.

Omkar Sanwatsarkar (24202103)

Omkar led the UI/UX design and overall visual coherence of the dashboard. He established a consistent design system in Tailwind, defining typography, spacing, and elevation, as well as a color palette linked to bloom risk levels. He ensured uniform styling across cards, charts, tables, and maps, while adapting layouts for smaller screens. Omkar also clarified labels, tooltips, and the information hierarchy so users could quickly move from headline risk levels to supporting evidence. He crafted UX copy for the LLM panel and empty/error states to reduce complexity and cognitive load for end users.

In addition to visual design, Omkar focused on aligning UX with performance. He instrumented logs for LLM response times and collaborated on prompt refinements to improve perceived latency and clarity. He also conducted informal usability walkthroughs, gathering feedback from different user groups, and converted this into targeted UI adjustments such as control placement, default filters, and legend clarity. Through his contributions, the dashboard was made coherent, accessible, and task-oriented across diverse user scenarios.